Griffiths 2021 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

67/95 samples (≥100 cells) intermix on one continuous manifold; per-neighbourhood iLISI(sample) 1.02→15.3 (θ=5), one kNN component (0 detached). The two density lobes are a proliferation / ribosomal axis (identical module scores), not per-patient islands.

Gate 2 — module geography → NO-GO

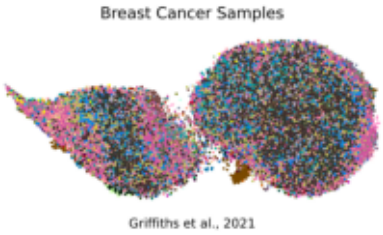
No Gast/pSom hub and no two-arm divergence: Gast-high and pSom-high populations are spatially anti-located (Jaccard 0.09; co-high fraction 0.007, below the 0.01 independence expectation) and stay negatively related after removing the ribosomal axis (partial r=−0.15) — Gast and pSom mark the two housekeeping lobes. The only embryonic relationship, a weak (Gast|pSom)–vs–Crest anti-correlation (r=−0.85), is transverse to the trajectory (|pseudotime ρ|≤0.11) — matches neither GO template. The embryonic pattern does not reproduce.

Gate 3 — cross-repression → NA

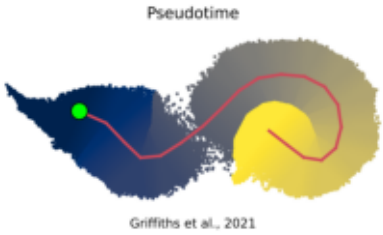
Not reached — Gate 2 NO-GO halts the chain, and the manifold is non-branching, so there is no hub or branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

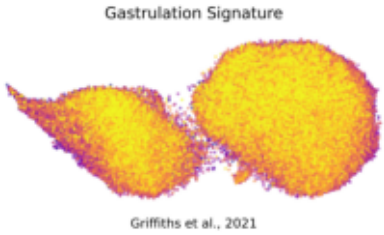
Manifold — samples



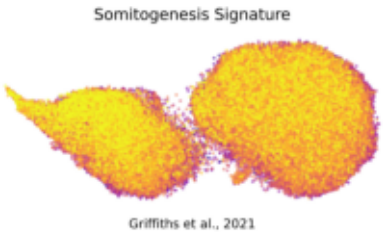
Pseudotime



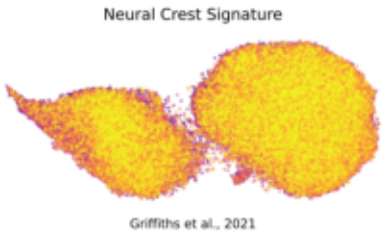
Gastrulation



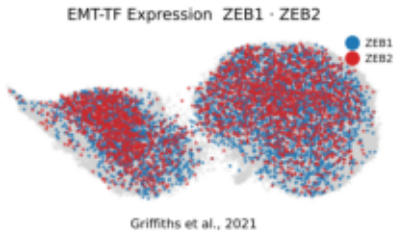
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Breast cancer (Griffiths et al., 2021), neoadjuvant letrozole+ribociclib trial, 10x + iCell8 — 41 patients / 111,372 malignant cells. 100-cell floor keeps 67 samples; after stripping 2,324 islands and scoping 5,821 cycling cells, 102,179 cells form one continuous ER+ luminal body carrying the (attempted) trajectory. Non-branching.

Verdict.

Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. The manifold is one integrated, continuous, non-branching luminal body, but its dominant geometry is a proliferation/ribosomal axis, not the embryonic program. There is no Gast/pSom hub and no two arms — Gast and pSom are anti-located, marking the two housekeeping lobes (co-high 0.007 < chance). The only embryonic relationship is a weak (Gast|pSom)–Crest anti-correlation (r=−0.85) transverse to the trajectory. Corrects five prior attempts that forced a Gate-2 GO on this housekeeping-confounded signal. Module means Gast −0.021 / pSom −0.052 / Crest +0.021.

Parameter choices

Min cells/sample	≥100 floor (67/95; smooth dist, no cliff)	**Integration**	Harmony θ=5 (sample); PCA20; spread=1.3
Scope	cycling + islands out; luminal body kept (102,179 cells)	**Trajectory**	scFates curve (ElPiGraph) N=20; 2 tips / 0 forks
Root node	curve tip, Gast/pSom-high Crest-low	**Geometry**	Linear / non-branching (curve 2 tips)
EMT-TFs	ZEB1 (~8.8%), ZEB2 (~5.0%) det.-gated; flat/non-directional; no VIM	**Overdispersed**	3,444 genes; no scaling